Upward and Downward Spirals in Employee Well-Being: Personality Profiles and Psychological Resources Under AI-Related Workplace Change

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This article reports secondary analyses of the LISS panel (Longitudinal Internet studies for the Social Sciences), administered by the non-profit research institute Centerdata (Tilburg University, the Netherlands). Centerdata granted access to the de-identified respondent files under a signed data-user agreement; respondent-level microdata are not redistributed. The analyses extend a prior latent personality–well-being profile program (Monnot, 2026). The design and analyses were not preregistered. As a secondary analysis of de-identified archive data collected and administered by Centerdata, the study involved no new data collection from human participants and did not constitute human-subjects research requiring institutional review board approval. Vocentive (San Francisco, CA, United States) is the institution responsible for this work. This research received no external funding, and the author has no conflicts of interest to

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Abstract

Artificial intelligence (AI) is changing how work is assigned, monitored, and rewarded. These changes tax the psychological resources that support employee well-being. Drawing on conservation of resources (COR) theory, this study examined latent personality profiles, AI-related stressors, and psychological resources as antecedents of employee well-being in the Dutch Longitudinal Internet studies for the Social Sciences (LISS) panel (2008–2025; N = 72,433 person-waves). Measurement-invariant latent profile models recovered Distressed, Antagonistic, Reserved, and Resilient configurations that corresponded to the four-profile solution found in previous research (mean domain r = .91; Tucker congruence = .96), with Distressed the partial shape match. Displacement risk and exhaustion measured in 2020 were associated with lower job and life satisfaction in 2025; positive affect was largely unrelated except under exhaustion, and Distressed-amplification moderation was not supported. Optimism, self-esteem, work meaning, and workplace relatedness related positively to evaluative well-being. Among adults leaving the Distressed profile, higher optimism, self-esteem, and work meaning were associated with recovery-oriented (Distressed to Resilient) rather than disengagement-oriented (Distressed to Reserved) exits, robust to classification-error sensitivities. Within-person cross-lags matched COR spiral predictions, and a random-intercept cross-lagged model confirmed a loss-primacy asymmetry for the self-esteem–distress pair. Perceived displacement, rather than occupation-level AI exposure, carried the psychosocial and distress associations. Distressed membership was also associated with more health-related lost work time and higher turnover intent than Resilient membership, although realized exits from employment did not differ. Psychological resources operate as caravan contents that distinguish recovery from disengagement during technological change.

Keywords: latent profile analysis, conservation of resources, technostress, displacement risk, employee well-being

Upward and Downward Spirals in Employee Well-Being: Personality Profiles and Psychological Resources Under AI-Related Workplace Change

The artificial intelligence (AI)-driven reorganization of work carries documented costs for employees. Industrial-robot intensity has been associated with worse mental health among manufacturing workers (Abeliansky et al., 2024). Employees who appraise their jobs as replaceable by smart technology, AI, robotics, and algorithms report lower commitment and higher turnover intentions (Brougham & Haar, 2018). U.S. regions more exposed to robots show more drug- and alcohol-related deaths even as workplace injuries fall (Gihleb et al., 2022). How employees maintain well-being under these conditions is a pressing question for organizational research. Nearly half of U.S. employment has been estimated to sit in occupations with a high probability of computerization (Frey & Osborne, 2017), and task-level measures of overlap with AI patents indicate that exposure now reaches high-skilled work (Webb, 2020). Industrial robots have been linked to lower local employment and wages (Acemoglu & Restrepo, 2020), and algorithmic management is changing how the remaining work is assigned, paced, and monitored (Kellogg et al., 2020), even as organizations weigh automating against augmenting human work (Raisch & Krakowski, 2021). For employees, these developments arrive as concrete demands: displacement threat operates as a core job-insecurity stressor (Shoss, 2017), and electronic performance monitoring is associated with higher stress without performance gains (Ravid et al., 2023). Technostress research links such information-technology demands to role stress, strain, and lower satisfaction and productivity (Ragu-Nathan et al., 2008; Tarafdar, Tu, et al., 2007), and distinguishes that threat-based techno-distress from challenge-based techno-eustress (Tarafdar, Cooper, & Stich, 2019). Occupation-linked exposure scores and survey measures of

displacement risk, machine pacing, and digital monitoring make this environment measurable (Felten et al., 2018, 2021), so its consequences for well-being can be tested rather than assumed.

Evidence from the panel studied here makes the concern concrete. In the Dutch Longitudinal Internet studies for the Social Sciences (LISS) panel, employees in occupations most susceptible to automation are the least likely to participate in work-related learning, a gap traced to learning intentions and perceived training opportunities rather than to perceived automation risk (Jansen et al., 2026). The present study treats this transition as a measured candidate mechanism rather than background context. Perceived displacement risk, exhaustion under technology-intensified work, algorithmic management, and technology self-efficacy, assessed in dedicated 2020 LISS modules, are tested as antecedents of later job satisfaction, life satisfaction, and positive affect.

Distressed Profiles and the Quality of Exits

Distressed personality configurations are identifiable, consequential, and capable of shifting over years, so well-being under technological strain cannot be reduced to mean trait levels alone. Person-centered mixture models recover multivariate Big Five prototypes rather than independent trait slopes, most often resilient, overcontrolled, and undercontrolled types (Asendorpf et al., 2001), with enumeration guided by fit, class size, and interpretability (Spurk et al., 2020; Woo et al., 2018). In LISS itself, prior latent-profile work retained a two-class resilient versus nonresilient solution (Asselmann et al., 2021). Four latent profiles, estimated from Big Five and agency indicators, have been recovered and tracked across midlife, and transition destination among adults leaving a Distressed configuration carried distinct implications: recovery-oriented movement toward a Resilient profile was associated with lower health-related lost productive time than disengagement-oriented movement toward a Reserved profile (Monnot,

2026). That four-profile solution, estimated in the Midlife in the United States (MIDUS) panel, supplies the reference configuration for the alignment tests reported here. The who-moves question among latent statuses is the estimand of latent transition analysis (LTA), complementing growth models that answer how much continuous distress or resource change accumulates (Nylund-Gibson, Garber, et al., 2023; Nylund-Gibson, Grimm, et al., 2014); the present study addresses that estimand with measurement-invariant profile models and empirical membership dynamics. Longitudinal LISS work already links resilient versus nonresilient membership to differential exposure and reaction to stressful life events, most notably a weaker symptom rise before unemployment and a stronger rebound after disability onset (Asselmann et al., 2021), and annual Personality waves permit short-gap membership dynamics that decade-spaced panels cannot resolve. If Distressed membership is sticky for some adults and permeable for others, then research on technological strain must model both profile structure and the quality of exits from distress.

Psychological Resources as Caravan Contents

Psychological resources are central to whether personality configurations remain integrated and generative or become depleted and withdrawn under sustained strain. Integrated functioning depends on personal evaluations and meaning structures that organize trait expression into coherent, approach-oriented action; when those resources erode, adults may reduce effort, interpersonal investment, and developmental engagement, the disengagement pattern that resource-based models predict under depletion (Demerouti et al., 2001; Hobfoll, 1989). Self-esteem indexes global self-worth and is tightly linked to emotional stability and adaptive coping (Judge et al., 2002; Rosenberg, 1965). Dispositional optimism, as measured by the Life Orientation Test-Revised (LOT-R), carries positive outcome expectancies that remain

associated with coping and lower depressive symptoms after neuroticism and related traits are controlled (Scheier et al., 1994). Work meaning, the experience of positive meaning and greater-good contribution at work, anchors purpose-connected engagement when roles change (Steger et al., 2012). Workplace relatedness, experienced as support and interpersonal connection on the job, supplies a social resource that travels with personal resources in conservation-of-resources and work–home resources accounts (Halbesleben et al., 2014; ten Brummelhuis & Bakker, 2012). Psychological capital (PsyCap), the composite of hope, efficacy, resilience, and optimism, functions as a state-like personal resource bundle predictive of satisfaction and performance (Luthans et al., 2007). These resources rarely operate in isolation. Conservation of resources theory describes them as traveling in resource caravans, aggregated packs of personal and social resources that are cultivated or depleted together by shared environmental conditions (Hobfoll et al., 2018). On this view, the resources examined here are candidate mechanisms of personality integration and of recovery- versus disengagement-oriented change, not incidental correlates of profile membership.

Conservation of resources (COR) theory supplies the organizing account of why these resources should matter for well-being under AI-related change. COR posits that people strive to obtain, retain, and protect valued resources, and that resource loss is disproportionately salient relative to equivalent gain (Hobfoll, 1989). Two corollaries carry the theoretical weight here. First, loss and gain are spiral processes: initial loss depletes the reserves needed to prevent further loss, whereas resource gain positions people to invest surplus resources and accrue further gain (Hobfoll et al., 2018). Second, resources travel in caravans that are shaped by caravan passageways, the environmental conditions that foster, protect, or block resource maintenance (Hobfoll et al., 2018). Job demands–resources research converges on the same

architecture: demands and technostress creators predict strain, whereas personal and job resources predict motivation and buffer impairment (Bakker & Demerouti, 2017; Demerouti et al., 2001). Under COR, AI-related stressors threaten resource reservoirs through replaceability threat, exhaustion, and loss of control under algorithmic pacing, whereas LOT-R optimism, self-esteem, work meaning, relatedness, and PsyCap are the caravan contents that should preserve well-being and tilt Distressed exits toward Resilient rather than Reserved destinations. The spiral corollary further implies within-person lagged associations between resources and distress, beyond stable between-person level differences (Hobfoll et al., 2018; see also Hamaker et al., 2015).

Overview of the Present Study

The present study tests one overarching claim: among Dutch employees facing measured technology-related resource threat, the personal and social resources that COR describes as a caravan (optimism, self-esteem, work meaning, workplace relatedness, and PsyCap; Hobfoll et al., 2018) covary with evaluative well-being and steer exits from the Distressed profile toward recovery rather than disengagement. The study was designed to contribute in three related ways. First, it tests whether the reference four-profile configuration generalizes across panels, instruments, and labor markets. Person-centered research requires this replication step before a profile solution is put to substantive use (Spurk et al., 2020; Woo et al., 2018). The test asks whether the profiles estimated in U.S. midlife adults (Monnot, 2026) can be recovered on a different Big Five instrument in a Dutch working-age panel and aligned by domain linking and orthogonal Procrustes rotation; membership stability is a separate, previously supported claim. Second, it tests the central COR predictions: that 2020 technology-related stressors show lagged associations with later well-being, that caravan resources covary with well-being, and that

resources are associated with recovery-oriented rather than disengagement-oriented exits from the Distressed profile (Hobfoll, 1989; Hobfoll et al., 2018). Third, it examines which operationalization of technological threat is associated with well-being costs, contrasting appraised displacement risk with occupation-level AI exposure scores (Felten et al., 2021; Shoss, 2017) and testing the within-person resource–distress lags that the spiral corollaries predict (Hamaker et al., 2015; Hobfoll et al., 2018). Measured stressors are 2020 robotization and technostress appraisals rather than generative AI. Machine-learning heterogeneous-effect estimators remain a secondary counterweight to these parametric tests (Chernozhukov et al., 2018; Wager & Athey, 2018).

Hypotheses

Hypothesis 1 (H1) concerns the measurement foundation and asks whether the reference profile configuration is recovered in LISS. Hypotheses 2 and 3 (H2, H3) concern the stressor side of the resource ledger, testing whether 2020 AI-related stressors show lagged associations with later well-being and whether Distressed membership amplifies them. Hypothesis 4 (H4) concerns the resource side, testing whether caravan contents steer the direction of Distressed exits; level associations between resources and well-being are reported as a descriptive validity check. Hypotheses 5 and 6 (H5, H6) probe the specificity of the COR account, first in within-person spiral dynamics and then in the dissociation between appraised and occupation-level exposure. Hypothesis 7 (H7) extends the resource ledger to its organizational bottom line, asking whether Distressed membership carries measurable employer-relevant costs.

Hypothesis tests that depend on an initial latent profile solution require that the solution itself be defensible before substantive predictions are evaluated, a two-study logic common in person-centered research (Spurk et al., 2020; Woo et al., 2018). Because the present predictions

import the Distressed–Resilient–Reserved destination logic from the reference panel, the LISS solution must first be shown to reproduce that configuration, not merely a four-class count. Domain-score linking and orthogonal Procrustes rotation supply the alignment tests. This configural correspondence test is distinct from the Series-1 membership-stability claim, which was supported previously.

H1. The LISS four-profile Big Five configuration will show strong domain-score linking to the matched MIDUS profiles (mean domain $r \geq .80$) and acceptable orthogonal Procrustes fit (Tucker congruence $\geq .90$; disparity $\leq .20$).

Under COR, the 2020 stressors are resource threats of distinct kinds. Displacement risk is the appraised threat of losing employment-linked resources; exhaustion indexes energy depletion already under way; algorithmic management removes control over how work is paced and evaluated (Hobfoll, 1989; Kellogg et al., 2020). Threatened and actual loss should carry forward into lower evaluative well-being. Technology self-efficacy, by contrast, is itself a personal resource aligned with the challenge-based techno-eustress pathway (Tarafdar, Cooper, & Stich, 2019).

H2. AI-related stressors indexing resource threat or loss (displacement risk, exhaustion, and algorithmic management) will be negatively related to later job satisfaction, life satisfaction, and positive affect; technology self-efficacy will be positively related to those criteria.

COR further predicts that loss begets loss: adults with depleted reservoirs are less able to buffer new threats, so the same stressor should cost more where resources are already scarce (Hobfoll et al., 2018). The Distressed profile is the configuration-level marker of that depletion. If loss spirals operate at the configural level, Distressed members should show steeper negative

stressor slopes than Resilient members. This a priori contrast is tested directly; if it is rejected, any stratified pattern among other profiles is treated as secondary, exploratory description only.

H3. Profile membership will moderate stressor–well-being associations such that stressor slopes are more strongly negative for Distressed members than for Resilient members.

Caravan contents show their most basic signature as positive associations with evaluative well-being, and primary studies and integrative reviews link each candidate resource to satisfaction criteria (Halbesleben et al., 2014; Judge et al., 2002; Luthans et al., 2007; Scheier et al., 1994; Steger et al., 2012). Those level associations are well established and are reported here as a descriptive validity check. The more stringent COR test concerns direction rather than level. Adults leave the Distressed profile along qualitatively distinct pathways, recovery toward the Resilient configuration or disengagement toward the Reserved one, and those destinations have carried distinct implications for capability and functioning (Monnot, 2026). Gain spirals imply that adults holding more caravan resources at the point of exit are positioned to reinvest and move toward recovery, whereas depleted movers should conserve what remains by withdrawing (Hobfoll et al., 2018).

H4. Among adults leaving the Distressed profile, higher resources will be associated with greater odds of a recovery-oriented transition to Resilient relative to a disengagement-oriented transition to Reserved.

Spiral corollaries are within-person claims: they concern whether a person's resource standing at one wave predicts that same person's distress at the next, over and above stable between-person differences (Hamaker et al., 2015; Hobfoll et al., 2018). Panel research has used reciprocal cross-lagged effects between resources and strain to study gain and loss processes (Hakanen et al., 2008), which requires that both constructs be measured repeatedly on the same

cadence with adequate reliability. Among the Series 2 resources, self-esteem and optimism are the only multi-item resource scales carried in the annual Personality core; work meaning and PsyCap are one-off modules, and relatedness is a single annual item conditioned on continuous employment. The spiral tests therefore pair self-esteem with mental-health distress and optimism with negative affect, the two resource–strain pairs with parallel annual multi-item measurement; this is a measurement-cadence restriction, not a claim that spiral dynamics are unique to these resources.

H5. Within persons, lagged self-esteem will predict lower subsequent mental-health distress, and lagged distress will predict lower subsequent self-esteem; lagged optimism and negative affect will show analogous reciprocal associations.

COR also locates threat in appraisal rather than in the occupation title. Perceived displacement risk is the appraised threat of resource loss, whereas occupation-level AI exposure scores index labor-market exposure that many incumbents may never appraise as threatening (Hobfoll, 1989). If the psychosocial costs run through appraisal, perceived risk should relate to work meaning, relatedness, and distress where occupation-level exposure does not.

H6. Perceived displacement risk will be negatively related to work meaning and relatedness and positively related to distress, whereas occupation-level AI exposure will show weaker or null associations with those psychosocial endpoints (domain dissociation).

Finally, if Distressed membership marks a depleted resource configuration, COR predicts that the depletion will surface in employer-relevant behavior, not in self-reported well-being alone: strained employees lose productive work time to health problems and orient toward withdrawal (Hobfoll et al., 2018). The costing literature supplies the translation. Depression-linked lost productive time carries substantial per-worker cost when hours are valued at the wage

(Stewart et al., 2003), work-related stress imposes large societal costs dominated by productivity losses (Hassard et al., 2018), and replacing a departed employee costs roughly one fifth of annual salary in typical jobs (Boushey & Glynn, 2012). Dutch practice adds the friction-cost correction, which caps absence costs at the period an employer needs to restore production (Koopmanschap et al., 1995). Together these methods permit an associational test of whether the Distressed configuration is more expensive to employ, on lost-time, turnover-intent, and advancement criteria.

H7. Distressed profile membership will be associated with greater employer-relevant costs than Resilient membership, indicated by more health-related lost productive time, higher turnover intent, and poorer promotion prospects.

Method

Transparency and Openness

This study analyzed existing LISS archive data. Centerdata granted access to the de-identified respondent files under a signed data-user agreement; respondent-level microdata are not redistributed. Because the study is a secondary analysis of de-identified archive data collected and administered by Centerdata, it involved no new data collection from human participants and did not constitute human-subjects research requiring institutional review board approval. The Appendix lists the data source, the files and modules used, other published studies that have used these data, and the author's own uses of the dataset. Analysis scripts that produce aggregate tables are maintained in the project repository. No preregistration was filed for this follow-up; estimands, sample rules, and hypothesis labels were documented in an analysis plan prior to compiling the Series 2 battery reported here. Hypothesis 7 was formulated after that plan

and is reported as exploratory. Supplemental materials report the resource–well-being validity-check paths, economic-cost sensitivity variants, and exploratory stratified slopes.

Participants and Procedure

Data came from the LISS panel, a probability-based household panel in the Netherlands (Scherpenzeel & Das, 2011). The structural backbone is the Core Personality study (50-item International Personality Item Pool [IPIP-50] Big Five and annual well-being batteries) across primary waves from 2008 to 2025, excluding Personality years with low IPIP completion (Figure 1). The measurement-invariant latent profile sample (with empirical membership dynamics) comprised 72,433 person-waves (16,862 unique persons) after excluding person-waves with intraindividual response variability below 0.25 across the 50 IPIP items (careless/invariant responding). Single-wave $k = 4$ solutions are uneven early in the panel (elevational splits; Distressed clearer later), so the stacked invariant model is primary; low-completion years are excluded because including them collapses Distressed in sensitivity checks. Hypothesis tests that join assembled modules use listwise overlap with paid-employment filters where work criteria are required. AI-related stressors were fielded primarily in 2020 (Module 232, Resilience towards robotization; Module 292, Survey Technostress). PsyCap was fielded in 2019 (Module 294). Work meaning was fielded in 2023 (Module 350). Annual Work and Health cores supply relatedness, job satisfaction, and mental-health distress. Because several predictors are module-timed, stressor → 2025 well-being models are lagged associations with temporal precedence, not within-person change designs.

Panel attrition and refreshment are material over 18 years and are reported directly (Figure S1 in the online supplemental materials). Among the 14,003 analysis persons entering the panel by 2020, 30.6% contributed Personality data again in 2023–2025 (median 3 waves per

person; 27.5% observed once). Baseline selectivity was modest: attrited persons scored lower on baseline conscientiousness ($d = -0.18$) and optimism ($d = -0.09$) and were slightly more often baseline Distressed (20.1% vs. 18.8%); remaining trait and resource differences were $|d| \leq 0.06$. Attrition of this kind leaves a modestly positively selected panel. Its expected consequence is restricted variance on distress-linked predictors, which attenuates associations rather than reversing their sign (Goodman & Blum, 1996). The stacked profile model uses all available person-waves; hypothesis tests are complete-case within their module overlaps.

Measures

Personality Profiles

Big Five scale scores were formed from the IPIP-50 (Goldberg, 1992; 10 items per trait; e.g., "Am the life of the party" for extraversion, "Get stressed out easily" for neuroticism; αs in recent waves ≈ .77–.90). Scores were standardized within the analysis sample. A measurement-invariant stacked diagonal Gaussian mixture model with 80 random starts retained a four-class solution (best log-likelihood replicated in 21 of 80 random starts; entropy = .537) labeled Distressed, Antagonistic, Reserved, and Resilient by shape-matching to MIDUS centroids (Asselmann et al., 2021; Monnot, 2026; Spurk et al., 2020). Retention rested on multi-criterion judgment (class size, interpretability, MIDUS shape match, and cross-wave coherence), not best log-likelihood replication alone. Distressed combines elevated neuroticism with preserved agreeableness; Antagonistic shows the lowest agreeableness and a depleted elevational pattern; Reserved is near average with mild withdrawal; Resilient shows high agency-relevant traits and low neuroticism. Longitudinal membership dynamics (adjacent- and multi-year stay/transition rates) were computed from modal assignments to the invariant solution; this implements the who-moves estimand of LTA without a simultaneous latent transition matrix in one step.

AI-Related Stressors (2020)

Perceived displacement risk was a multi-item composite from Module 232 (replaceability by AI/robots/algorithms and related worry items; e.g., “My occupation could be performed by new technologies [for instance artificial intelligence, robots or algorithms]”). Exhaustion and algorithmic-management indicators came from Module 292 (machine- or manager-paced work, digital output registration, customer-rating appraisal, and exhaustion under technology-intensified conditions; e.g., work pace set by “the automatic speed of a machine or movement of a product”; “my work activities have become mentally more exhausting in the past five years”). These map onto technostress creators (e.g., overload/invasion-adjacent strain) and algorithmic control (Kellogg et al., 2020; Tarafdar, Cooper, & Stich, 2019; Tarafdar, Tu, et al., 2007), which helps explain why exhaustion shows broader evaluative associations than algorithmic management alone. Technology self-efficacy indexed confidence in using new technologies (e.g., “I am well able to work with the new technologies required by my occupation”). Where used, AI Occupational Exposure (AIOE) scores (Felten et al., 2021), which extend the ability-link method of Felten et al. (2018), were crosswalked from U.S. Standard Occupational Classification codes to International Standard Classification of Occupations (2008 revision) codes and merged on occupation.

Well-Being Criteria

Job satisfaction was measured with the annual Work-core rating (0–10) and a 4-point Work item. Life satisfaction used the five-item Satisfaction With Life Scale (SWLS; Diener et al., 1985; e.g., “In most ways my life is close to my ideal”; $\alpha \approx .90$). Positive and negative affect used the Positive and Negative Affect Schedule state scales (Watson et al., 1988; e.g., “enthusiastic” and “distressed”; αs $\approx .89$ and .94). Mental-health distress used the five-item

Mental Health Inventory (MHI-5; Berwick et al., 1991; e.g., “I felt so down that nothing could cheer me up”; α ≈ .87).

Psychological Resources

Optimism was the LOT-R scored composite (Scheier et al., 1994; e.g., “In uncertain times, I usually expect the best”; α ≈ .76). Self-esteem was the 10-item Rosenberg scale (Rosenberg, 1965; e.g., “On the whole, I am satisfied with myself”; α ≈ .91). Work meaning was a 10-item Module 350 battery (e.g., “I understand how my work contributes to my life’s meaning”; α ≈ .92). Workplace relatedness was the annual Work-core support item (“sufficient support in difficult situations”), treated as a single-indicator social resource rather than a multi-item need-satisfaction scale. PsyCap was the 12-item Compound PsyCap Scale total and efficacy facet from Module 294 (CPC-12; Lorenz et al., 2016; e.g., “I am confident in my ability to deal effectively with unexpected situations”; α = .89), measuring the PsyCap construct of Luthans et al. (2007). The CPC-12 is a validated domain-general compound measure rather than the work-referenced Psychological Capital Questionnaire, and Module 294 administered it with an explicit health-behavior frame (respondents were instructed to think about how their own behavior might affect their health) in a subsample of 600 panel members stratified by socioeconomic status (SES; 479 respondents; fielded February 2019). Multi-item basic psychological need batteries are not available as annual single-construct scales in this build and are not used as primary Series 2 predictors.

Economic-Cost Criteria (2023–2025)

Health-related lost productive time (LPT) combined annual Health-core reports of days unable to work, days of partial work, and the degree to which health interfered with work (categorical responses converted to day and fraction midpoints), annualized and valued at each

respondent's hourly net wage under the human-capital method (Stewart et al., 2003). Sensitivities comprised a Dutch friction-cost variant capping absence at the 85-day friction period of the Dutch costing manual (Kanters et al., 2017), following the friction-cost method (Koopmanschap et al., 1995), and a loaded-wage variant (× 1.35). Turnover intent used the 2023 Module 350 quit-likelihood item (likely or very likely to quit within 12 months) and, as a scenario translation, the expected replacement cost of an intended exit at 21% of annual salary (Boushey & Glynn, 2012). Realized exits from paid employment (2023 → 2025), self-rated effort (0%–100%), and the Work-core promotion-prospects item (1–4; higher = poorer prospects) completed the criterion set. Euro values are associational burden translations under stated multipliers, not measured firm outcomes.

Analytic Strategy

Series 2 tests were estimated on disclosure-safe analysis frames. Continuous outcomes used standardized ordinary least squares (OLS) regression with heteroskedasticity-consistent (HC3) standard errors and demographic covariates (age, gender, and education) where samples allowed. Education entered as the ordered LISS diploma code (1 = primary through 6 = university); the codes for other, not completed, and not yet started education (about 2% of paid employees) were set to missing rather than ranked. Regression models required at least 80 complete cases (120 for the product-term moderation models; 40 for the leaver-destination logits). Unless identified as one-tailed (the directional loss-primacy contrasts), tests were two-tailed at $\alpha = .05$. Within the H2 stressor family and the resource–well-being validity-check family, Benjamini–Hochberg false-discovery-rate control ($q < .05$) was applied as multiplicity hygiene. Profile moderation (H3) used product-term models (Reserved reference) and stratified simple slopes on modal assignments, without correction for classification error.

Hypothesis 1 compared matched LISS and MIDUS profile means via (a) within-domain correlations across the four profiles and (b) orthogonal Procrustes rotation of the profile × trait matrix, reporting Tucker congruence and disparity. As an internal replication exploiting LISS recruitment refreshment, profiles were also re-enumerated independently within panel-entry cohorts (2008–2009, 2011–2017, and 2019–2025) and compared with the invariant centroids via Tucker congruence. Retention decisions followed established multi-criterion guidance (Spurk et al., 2020): information criteria often continued to favor more classes, so $k = 4$ was retained for MIDUS-comparable configural structure, class size, and interpretability rather than the Bayesian information criterion alone. Soft posterior responsibilities (mean γ) are reported as a Gaussian-mixture counterweight to hard shares and do not replace mixture or COR tests. As a cross-software check, the profile model and a three-wave latent transition model with measurement fixed at the invariant solution were re-estimated in Mplus 8.8 on the same person-level frames (Muthén & Muthén, 2017).

Logistic models for Distressed leavers (Hypothesis 4) contrasted destination Resilient versus Reserved, the recovery and disengagement pathways identified in previous research (Monnot, 2026). Because distal profile tests use modal assignments at entropy = .537, Hypothesis 4 was re-estimated under three classification-error sensitivities: (a) soft-weighted destination logits in which each leaver was weighted by the posterior probability of origin-wave Distressed membership multiplied by the summed posterior probabilities of the two destination profiles, with fractional destination probability as the outcome; (b) 200 pseudo-class draws from the posterior class probabilities at both waves; and (c) Firth bias-reduced logistic regression (Firth, 1993) on the modal leavers, which stabilizes the small PsyCap overlap. The soft-weighted and pseudo-class estimators follow the three-step auxiliary-variable literature (Asparouhov &

Muthén, 2014; Nylund-Gibson, Grimm, et al., 2014). The transition covariate was then estimated inside the latent transition model itself, with the misclassification matrix fixed from the invariant solution's posteriors (three-step covariate estimation; Vermunt, 2010). With a single transition pair per person at entropy = .537, the profile likelihood for the transition slope proved empirically flat: covariate-dependent origin membership combined with class-specific transition intercepts reproduces any covariate-varying destination pattern, so origin selection and transition steering (which names a model component, not a causal effect) are separately identified in these data only from longer sequences. The identified sequence-based estimate was therefore a latent annual Markov chain over the four profiles (2008–2025 grid; panel gaps bridged by repeated annual transitions under a time-homogeneous annual transition assumption), with emissions through the fixed misclassification matrix, person-mean resources on the initial-state logits, and the focal resource slope on the annual Distressed → Resilient versus Reserved transition logit, tested by likelihood ratio with profile-likelihood confidence intervals.

Within-person Hypothesis 5 used person-mean-centered lagged predictors with lagged dependent variables (Hamaker et al., 2015); loss-primacy asymmetry was tested by standardizing within persons and comparing the distress → resource and resource → distress cross-lags with a person-bootstrap one-tailed contrast. The spiral conclusions were then corroborated with three simultaneous-equation estimators: a random-intercept cross-lagged panel model estimated by full-information maximum likelihood via the Kalman filter, with random intercepts carried as constant states (Hamaker et al., 2015); a latent bivariate autoregressive model with free observation-noise variances that separates measurement error from within-person dynamics; and a person-level linear-detrending sensitivity in the spirit of latent curve models with structured residuals (Curran et al., 2014). Cross-lag and asymmetry claims in these models were tested by

likelihood ratio. The random-intercept specification was retained as primary because systematic person-specific growth was not hypothesized for these trait-adjacent annual series; the structured-residual growth model with slope variance fixed to zero is a special case of the random-intercept model (Usami et al., 2019), and the detrending sensitivity probes the consequence of relaxing that constraint without asserting a trajectory model.

Pandemic-period confounding was addressed descriptively and by a formal discontinuity check rather than by covariate adjustment. Profile prevalences show no abrupt 2020 break, and within-person well-being deviations around 2020 are small (Figure 2). A piecewise mixed-effects growth model with a 2020 knot (pre-2020 slope, 2020 level step, post-2020 slope; random intercept and post-knot slope) found no distress discontinuity, and LISS-based prospective work found no evidence that the pandemic altered personality–mental-health associations in this panel (van der Velden et al., 2024; cf. Vaziri et al., 2020, for latent transition designs spanning the COVID-19 onset). The same growth model, with occupation-level exposure interacting with the pre-slope, step, and post-slope terms, and a secondary heterogeneous-effect sweep (prespecified resource × exposure interactions under family false-discovery-rate control, with a double/debiased forest counterweight), extended the Hypothesis 6 dissociation to trajectories and to who-is-affected questions; the exposure × pre-slope term provides a pre-trend falsification check for occupational selection.

Hypothesis 7 contrasted Distressed membership against all other profiles and against Resilient alone on each economic criterion (OLS with HC3 standard errors on modal assignments, age-adjusted in the 2025 frame); frames differ by criterion (2025 profile × concurrent LPT and promotion prospects; 2023 profile × concurrent quit intent; 2023 profile → 2025 realized exits), and component euros are not summed across frames. Wage weighting used

monthly net income, which incorporates the LISS category-average imputation when an exact amount was refused; income nonresponse codes were treated as missing, and hourly wages required positive actual or, failing that, contract weekly hours. As the machine-learning counterweight for Hypothesis 7, a double/debiased machine learning pass with cross-fitted random-forest nuisances and an R-learner forest for conditional associations (Chernozhukov et al., 2018; Nie & Wager, 2021; Wager & Athey, 2018) re-estimated the Distressed contrasts on LPT euros and quit intent after flexible adjustment for age, sex, weekly hours, and wage, with a best-linear-predictor test for heterogeneity. These estimates pressure-test the classical contrasts and do not replace them.

Results

Descriptive Profile Structure

Under the invariant four-class solution, modal profile shares were Distressed 16.6%, Antagonistic 28.8%, Reserved 36.3%, and Resilient 18.3% (Table 1; Figure 3, Panel A). Mean soft posterior responsibilities (γ) were close to the hard shares (Distressed 19.4%, Antagonistic 31.1%, Reserved 31.9%, and Resilient 17.6%), indicating that modal (maximum a posteriori) assignment does not invent the minority Distressed and Resilient poles at entropy = .537. Adjacent-wave same-class probabilities averaged about .65–.71 (Figure 4); 10-year same-class membership remained a majority (about .58–.61). Distressed membership showed more decade-scale outflow than Antagonistic membership (Figure 3, Panel B). Calendar-year prevalences drifted slowly (Antagonistic +5 percentage points; Reserved −4 percentage points from 2008 to 2025) without an abrupt 2020 break (Figure 5).

Cross-software re-estimation in Mplus 8.8 reproduced this structure. Independent maximum-likelihood estimation recovered the four-class solution with matched class means (*r*s

≥ .999; largest absolute mean difference 0.034), model-based shares within 1 percentage point, and entropy = .541 versus .537. In a three-wave (2023–2025) latent transition re-estimation with measurement fixed at the invariant solution, Mplus and an equivalent Python expectation-maximization refit converged to the same log-likelihood and to an identity latent transition matrix, indicating that short-window modal movement is absorbed by classification error once measurement is fixed. Adjacent-wave modal probabilities therefore describe assignment dynamics rather than error-free latent change, and genuine profile movement is a longer-horizon phenomenon, consistent with the sequence-based estimates reported under Hypothesis 4.

Hypothesis 1: Configural Correspondence

Hypothesis 1 was supported at the stated mean-domain and Procrustes criteria. Matched one-to-one profile labels yielded domain-score linking correlations of .78 (neuroticism), .93 (extraversion), .92 (openness), .99 (agreeableness), and .94 (conscientiousness; mean r = .91). Neuroticism linking alone fell slightly below .80 while the mean criterion was met. After orthogonal Procrustes alignment, Tucker congruence (φ) was .96 and disparity was .09 (Table 2). Shape correlations were strongest for Antagonistic (r = .99) and Resilient (r = .96); Distressed (r = .69) and Reserved (r = .74) were weaker but directionally correct. Distressed remains the partial configural match, combining high neuroticism with high agreeableness, which is consistent with instrument and extremity differences between Midlife Development Inventory (MIDI) factor scores and IPIP scale means.

Recruitment-cohort re-enumeration qualified this correspondence internally. Within each panel-entry cohort (2008–2009, 2011–2017, and 2019–2025), the Bayesian information criterion preferred four or more classes, and the Antagonistic and Resilient anchors re-emerged with strong congruence to the invariant centroids (best φ = .95–.97 and .88–.97, respectively). The

Reserved majority re-emerged cleanly in the most recent cohort (φ = .87) but split along elevation in the two earlier cohorts. The Agreeable-Sensitive Distressed configuration did not re-emerge within any single cohort (best φ = .50–.69); its recovery requires the pooled measurement-invariant model. This bounds the configural claim: the profile anchors replicate across independently recruited cohorts, whereas the focal Distressed configuration is a pooled-model object, consistent with its partial MIDUS shape match and its sensitivity to low-completion waves.

Hypothesis 2: AI-Related Stressors and Well-Being

Hypothesis 2 was partially supported (10 of 16 focal paths were significant, and all 10 survived Benjamini–Hochberg control; Table 3; Figure 6). These are lagged associations of 2020 stressors with 2025 outcomes (primary job-satisfaction criterion: 0–10 Work-core rating; 4-point item reported as robustness), not within-person change models. Displacement risk was associated with lower later job satisfaction (β = −.11, $p < .001$, $n = 918$) and life satisfaction (β = −.11, $p < .001$, $n = 1{,}131$) but not positive affect (β = .04, *ns*). Exhaustion showed the broadest pattern across evaluative criteria and the sole consistent negative association with positive affect (job satisfaction β = −.22; SWLS β = −.31; positive affect β = −.20; all $p < .001$). Algorithmic management was associated with lower life satisfaction (β = −.10, $p < .001$) but not job satisfaction or positive affect. Technology self-efficacy was associated with higher job and life satisfaction (βs ≈ .10) but not positive affect; its 4-point job-satisfaction robustness path fell just short of significance (β = .07, $p = .051$). Lagged AI-stressor associations were therefore primarily with evaluative criteria; positive affect was largely criterion-null except under exhaustion. Overlap sample sizes differ by employment filter (job-satisfaction paths ≈ 790–920 versus larger SWLS and positive-affect overlaps).

Hypothesis 3: Profile Moderation

Hypothesis 3 was not supported in the a priori Distressed-amplification / Resilient-buffer form: formal Distressed × stressor interactions versus Reserved were largely null, and pattern checks that stressor slopes are more negative for Distressed than for Resilient members succeeded in only a minority of comparisons. Exploratory stratified slopes instead identified Antagonistic membership as the configuration with the steepest negative stressor slopes (e.g., displacement → job satisfaction $\beta = -.25$ vs. $-.08$ for Resilient); this secondary description appears in the online supplemental materials (Table S1) and does not rehabilitate H3 as stated.

Validity Check: Resources and Well-Being

Resources behaved as COR predicts in direct models. Annual LOT-R optimism, self-esteem, work meaning, and workplace relatedness were each positively associated with job satisfaction, life satisfaction, and positive affect (standardized βs typically .12–.53, all $p < .001$; ns ≈ 1,480–4,500; 18 of 19 paths, all survivors of Benjamini–Hochberg control at $q < .05$), and PsyCap showed the same directional pattern at smaller module overlap (βs ≈ .32–.45; ns ≈ 240; the job-satisfaction overlap, $n = 79$, fell below the 80-case estimation floor), treated as converging, underpowered corroboration. Full path estimates appear in the online supplemental materials (Table S2).

Hypothesis 4: Resources and Recovery Versus Disengagement

Hypothesis 4 was partially supported for annually and module-measured resources on the MIDUS-congruent destination contrast (Table 4; Figure 7). Pooled-leaver interval odds ratios (ORs) are the primary Hypothesis 4 estimates; the sequence-based estimates reported below are identified corroboration on an annual, per-*SD*-of-person-mean scale and are not comparable in magnitude. Among pooled unique Distressed leavers, higher optimism ($OR = 1.78$, $p < .001$, n =

674), self-esteem (*OR* = 1.62, $p < .001$, $n = 797$), work meaning (*OR* = 1.77, $p < .001$, $n = 323$), and relatedness (*OR* = 1.34, $p = .012$, $n = 332$) were associated with greater odds of transitioning to Resilient rather than Reserved. PsyCap total was directionally consistent but not supported at the available sample size (modal *OR* = 1.41, $p = .29$, $n = 44$; Firth *OR* = 1.36, $p = .33$). Interval-frame classification-error sensitivities bound the entropy caveat: soft-weighted and pseudo-class (200 draws) odds ratios for optimism, self-esteem, and work meaning remained above 1 with draw intervals entirely above 1, whereas relatedness did not survive (soft *OR* = 1.14, $p = .227$; pseudo-class 95% interval [0.99, 1.36]); per-resource sensitivity values appear in Table 4. PsyCap passed the same interval sensitivities numerically (soft *OR* = 2.51, $p = .030$; pseudo-class median 2.05, interval [1.20, 5.19]); given the small leaver overlap and the null sequence estimate below, these passes are reported for symmetry and are not treated as rehabilitating the underpowered modal path.

On the corroborating annual latent scale, destination associations survived direct classification-error correction. The sequence-based three-step estimate (fixed misclassification matrix; full assignment sequences) yielded annual latent Distressed → Resilient versus Reserved odds ratios of 2.40 for optimism (95% profile CI [1.88, 3.13], $p < .001$, $n = 10{,}649$ persons), 3.72 for self-esteem ([2.62, 5.55], $p < .001$, $n = 12{,}232$), and 1.68 for work meaning ([1.14, 2.63], $p = .007$, $n = 2{,}429$). Samples are full assignment sequences for the panel, not the leaver subset. These are classification-error-robust associations, not causal effects: the person-mean resource proxies stable trait standing computed over the full observation window, so the slopes carry no temporal precedence and do not separate resource effects from stable-trait sorting. Relatedness, fragile in the interval frame, showed a reliable association on the trait-scaled sequence estimate (*OR* = 1.64, [1.26, 2.15], $p < .001$, $n = 5{,}602$). The sequence frame also substitutes a person-

mean of the single support item, a more reliable, trait-scaled predictor without the origin-year measure's temporal precedence, so the frames differ in estimand as well as in classification-error handling. Under the uniform reconciliation rule that the identified sequence estimator characterizes paths while interval sensitivities characterize fragility, the social-resource path is described as frame-sensitive rather than absent, and PsyCap, null in the sequence frame (*OR* = 1.61, [0.59, 4.14], $p = .32$, $n = 464$), is not recharacterized. Fielding windows differ: PsyCap was measured in 2019, work meaning in 2023, and optimism, self-esteem, and relatedness annually, while leavers are pooled across intervals, so meaning and PsyCap pathway estimates are especially order-sensitive and are not interpreted as synchronized prospective prediction. Multiplicative interactions between resources and aggregate work-support composites were not supported and are not primary evidence (relatedness remains a single-item social resource; basic psychological need batteries are not the theory spine).

Hypothesis 5: Within-Person Resource–Distress Spirals

Hypothesis 5 was supported (Table 5). Within-person lagged self-esteem was associated with lower subsequent MHI-5 distress ($\beta = -.03$, $p < .001$, $n = 7{,}477$ persons), and lagged distress with lower subsequent self-esteem ($\beta = -.12$, $p < .001$, $n = 6{,}780$). Lagged optimism was associated with lower subsequent negative affect ($\beta = -.06$, $p < .001$, $n = 6{,}748$), and lagged negative affect with lower subsequent optimism ($\beta = -.02$, $p < .001$, $n = 7{,}432$). After within-person standardization of the demeaned series, a person-bootstrap lagged asymmetry contrast tested whether the absolute distress → resource cross-lag exceeds the absolute resource → distress cross-lag. This contrast is consistent with loss primacy, although it does not compare loss against an equivalent gain in the same resource. The asymmetry was supported for both pairs (self-esteem ↔ MHI-5: asymmetry contrast $\delta = -.050$, 95% CI [−.071, −.030], one-tailed $p <$

.001; optimism ↔ negative affect (NA): δ = −.022, 95% CI [−.040, −.000], upper bound marginally below zero, one-tailed p = .025; 400 bootstrap replications). Differential reliability runs against the self-esteem–distress asymmetry (the dominant distress → resource path uses the less reliable MHI-5 predictor, α ≈ .87 vs. .91), so that contrast is robust to the attenuation artifact; it runs with the optimism–negative-affect asymmetry (LOT-R α ≈ .76 vs. negative affect α ≈ .94), so that contrast is fragile because attenuation of the less reliable optimism predictor could manufacture part of it. These within-person lagged associations are consistent with COR gain and loss spiral predictions after separating stable between-person differences (Hamaker et al., 2015; Hobfoll et al., 2018); they are not interventional or full spiral-process identification.

The simultaneous-equation corroboration qualified these conclusions (Figure 8). For the self-esteem–distress pair, the random-intercept cross-lagged panel model reproduced both within-person paths (distress → self-esteem β = −.14; self-esteem → distress β = −.08; likelihood-ratio ps < .001; n = 7,513 persons, 2017–2025), and the loss-primacy asymmetry was significant by likelihood-ratio test (p < .001), upgrading the bootstrap contrast to a formal simultaneous test. The latent autoregressive model with free observation noise strengthened rather than weakened both paths (−.22 and −.12; estimated observation-noise variances of 0.16 and 0.20 in pooled-*SD* units) while preserving the asymmetry direction (p = .080), indicating that the fixed-effects estimates are attenuation-bounded rather than artifacts of noisy single-scale measurement. For the optimism–negative-affect pair, the random-intercept model reproduced the reciprocal paths (−.06 and −.05, ps < .001; n = 7,470) but not the asymmetry (p = .335), and the free-noise variant was weakly identified (the optimism autoregression approached the random-intercept boundary), so only the random-intercept estimates are reported for that pair. A person-level linear-detrending sensitivity retained the distress → resource paths (self-esteem: β = −.04, p

< .001; optimism: β = −.03, p < .001) while the resource → distress paths fell to nonsignificance, a pattern in which gains ride slow person-level trends whereas losses register as abrupt, discrete changes, again consistent with loss primacy.

Hypothesis 6: Domain Dissociation

Hypothesis 6 was partially supported and hinged mainly on perceived displacement rather than occupation-level AIOE. Displacement risk was associated with lower work meaning (β = −.07, p = .025, n = 979), lower relatedness (β = −.08, p = .028, n = 931), lower job satisfaction (β = −.11, p = .002, n = 837), and higher MHI-5 distress (β = .16, p < .001, n = 1,032). AIOE was null on relatedness, job satisfaction, and distress; its one nominally significant path, to work meaning (β = −.07, p = .044), was small, was not prespecified for multiplicity control, and is treated as suggestive. The dissociation object is therefore primarily perceived replaceability versus occupational exposure scores; it is not a claim that job attainment is generally preserved under AI threat, because job satisfaction falls under displacement while AIOE is null on that criterion.

Trajectory and heterogeneity checks corroborated the dissociation (Figure 9). In the piecewise growth model, the population showed no 2020 distress discontinuity (level step = 0.02, p = .059; slopes gently declining on both sides of the knot) and a small positive life-satisfaction step (+0.06 on the 1–7 scale, p < .001). Occupation-level exposure did not moderate post-2020 distress trajectories on any score (no exposure × step or × post term survived family false-discovery-rate control). The one clean moderation was AIOE × step on life satisfaction (+0.03, p = .014, q = .06): higher-exposure occupations, which skew professional, gained slightly more life satisfaction at 2020 while their exposure × pre-slope falsification term was null. Frey–Osborne automatability failed that falsification check (exposure × pre-slope p < .001), indicating

occupational selection rather than pandemic-era disruption, so its knot terms are not interpreted. In the heterogeneity sweep, none of the 18 prespecified resource × exposure interactions on 2025 criteria survived family false-discovery-rate control; the two nominally significant terms were buffering-consistent Frey–Osborne × self-esteem and × optimism interactions on distress (*b*s ≈ −0.03, *p*s = .027 and .050, *q* = .075), and the forest counterweight reproduced the null average AIOE–distress association with only weak, nonmonotone heterogeneity. Occupation-level exposure therefore carries occupational composition rather than disruption, and no resource-defined subgroup shows a reliable exposure-linked well-being cost.

Hypothesis 7: Economic Costs of Distressed Membership

Hypothesis 7 was partially supported, with a domain split between lost time and intent on one side and realized exits and effort on the other (Table 6). Distressed members reported more annual LPT hours than Resilient members (+65 hr per year, $p = .020$, $n = 1{,}851$; vs. all other profiles pooled, +39 hr, $p = .078$) and more absentee days (+0.60 vs. Resilient, $p = .011$). Valued at the wage, the human-capital LPT differential against Resilient was +€999 per year but did not reach conventional significance ($p = .085$; vs. all other profiles pooled, +€691, $p = .114$), whereas the friction-cost variant remained reliable against Resilient (+€963, $p = .020$); Antagonistic members carried mean LPT euros comparable to Distressed members. On the intent criteria, 17% of Distressed members versus 8% of Resilient members reported being likely to quit within 12 months ($p < .001$), an expected replacement-cost differential of +€330 per year under the 21%-of-salary scenario, and Distressed members rated their promotion prospects as poorer (+0.25 scale points, $p < .001$). The cost account did not extend to realized behavior or to effort: exits from paid employment by 2025 did not differ (9% of Distressed vs. 9% of Resilient members, $p = .937$; vs. all other profiles pooled, +1 percentage point, $p = .445$), and effort-based

translations reversed because Distressed members reported more self-rated effort than every other profile (74.5% vs. 65.9%, $p < .001$). Distressed membership is therefore associated with a lost-time and withdrawal-intent burden, not a realized-turnover or low-effort burden; euro magnitudes are scenario translations sensitive to wage weighting, and wage-sensitivity variants (friction-cost capping, loaded wages, and realized-exit scenarios) appear in the online supplemental materials (Table S3).

The double/debiased machine learning counterweight reinforced this division: the quit-intent contrast survived flexible adjustment for age, sex, weekly hours, and wage (partialled Distressed association +.05, $SE = 0.02$, $p = .037$, $n = 2{,}031$), whereas the LPT-euro contrast attenuated to nonsignificance (+€405, $p = .35$, $n = 1{,}743$), with wage and age dominating the conditional-association structure. Raw-hours contrasts, which do not inherit wage weighting, remain the cleaner lost-time evidence. Best-linear-predictor heterogeneity tests were null ($ps \geq .45$). The classical contrasts remain the headline estimates; the counterweight bounds the euro-valued claims as partly compositional.

Discussion

This study asked whether the four-profile personality configuration found in previous research (Monnot, 2026) is recovered in LISS under geometric tests, whether 2020 technology-related stressors show lagged associations with later evaluative well-being, and whether COR-relevant resources covary with well-being and with recovery-oriented exits from Distressed membership. Configural correspondence held overall (H1 supported), with Distressed the partial shape match. Displacement risk and exhaustion were associated with lower later job and life satisfaction (H2 partially supported); positive affect was largely criterion-null except under exhaustion. A priori Distressed-amplification moderation was not supported (H3); exploratory

stratified slopes often showed steeper negative associations among Antagonistic members. Optimism, self-esteem, work meaning, and relatedness behaved as COR predicts in direct models (descriptive validity check). Optimism, self-esteem, and work meaning retained Distressed-leaver destination associations under interval-frame sensitivities and under direct classification-error correction on the sequence scale. Relatedness was fragile in the interval frame but reliable on the trait-scaled sequence estimate, and PsyCap remained underpowered (H4 partially supported). Within-person lags, corroborated by random-intercept cross-lagged and measurement-error-separating models with a likelihood-ratio-confirmed loss-primacy asymmetry for the self-esteem–distress pair (absent for the optimism–negative-affect pair), were consistent with the reciprocal resource–distress associations that gain and loss spirals predict (H5 supported). Hypothesis 6 was partially supported: perceived displacement, far more than occupation-level AI exposure, carried the psychosocial and distress associations. Extending the ledger to employer-relevant criteria, Distressed membership carried more health-related lost work time and higher turnover intent than Resilient membership, but not more realized exits or lower self-rated effort (H7 partially supported), and the euro-valued differentials proved partly compositional under the machine-learning counterweight.

Implications

For theory, the results reinforce treating transition quality, not distress prevalence alone, as a primary longitudinal outcome (Monnot, 2026). Latent transition analysis supplies the estimand for movement among statuses (Nylund-Gibson, Garber, et al., 2023; see also Chawla et al., 2020, on recovery-experience profiles as adjacent evidence). COR remains a productive bridge between person-centered profiles and technology-related strain, and the evidence maps onto its two central corollaries. The convergence of the resource–well-being validity check and

Hypothesis 4 is consistent with, though not a test of, the caravan principle: the same personal and social resources that covary with evaluative well-being also covary with recovery-oriented rather than disengagement-oriented exits. Each estimate is a separate single-resource model; inter-resource covariation and joint resource models were not estimated, so co-travel is inferred from convergence rather than demonstrated (Halbesleben et al., 2014; Hobfoll et al., 2018). The spiral corollary is visible in the within-person reciprocal lags of Hypothesis 5, including a lagged asymmetry consistent with the distress → resource dominance that loss primacy predicts; the asymmetry was significant by likelihood-ratio test in the random-intercept cross-lagged model for the self-esteem–distress pair and absent for the optimism–negative-affect pair. Partial configural correspondence (weaker Distressed shape match; entropy = .537) bounds how strongly LISS reproduces the reference solution.

For organizations monitoring technological strain, optimism, self-worth, work meaning, relatedness, exhaustion, and replaceability appraisals are markers that covary with evaluative well-being and with recovery-oriented Distressed exits. In COR's heuristic terms, work environments operate as caravan passageways that foster or block resource maintenance, but whether intervening on these targets improves outcomes is not identified here (Hobfoll et al., 2018; Luthans et al., 2007). Algorithmic management showed narrower evaluative associations than exhaustion and displacement, cautioning against treating all technology exposures as interchangeable (Kellogg et al., 2020; Tarafdar, Cooper, & Stich, 2019). The same caution applies to occupational exposure scores themselves: in the trajectory checks, Felten AIOE passed its pre-trend falsification test where Frey–Osborne automatability failed it, so conclusions that pool exposure metrics interchangeably risk mistaking occupational selection for technology effects (Felten et al., 2021; Frey & Osborne, 2017). The Hypothesis 7 pattern gives that

monitoring case an economic anchor: Distressed employees reported 39 to 65 more lost work hours per year than other profiles and more than twice the Resilient rate of quit intent (17% vs. 8%), and these costs preceded any realized exit. Work-related stress imposes societal costs dominated by productivity losses rather than turnover itself (Hassard et al., 2018), so screening and resource support targeted at distressed employees can begin before separation decisions are made, although no intervention effect is estimated here. Finally, one implication reaches back to the program that motivated this study. Prior analyses modeled how configurations, distress, and resources relate to functioning, and deferred the claim that technological change produces distress to designs pairing measured exposure with repeated assessment (Monnot, 2026). The present study executes the first step of that agenda: exposure is measured and temporally precedes the criteria, although single-shot 2020 modules still fall short of the repeated exposure assessment that a full mechanism claim requires.

Limitations and Future Directions

Several limitations bound inference. First, the design is observational: stressors and resources were not manipulated, so the findings cannot license the causal claims that random assignment affords. The multi-wave structure nonetheless supports stronger inference than cross-sectional designs in two specific ways. In the Hypothesis 2 lagged models, stressors were measured at waves preceding the criteria, establishing temporal precedence of measurement, although those models adjust for demographics and not for baseline criterion levels. In the Hypothesis 5 within-person models, person-mean centering removes stable individual differences, separating within-person change from the between-person differences that would otherwise confound it (Hamaker et al., 2015; see Ployhart & Vandenberg, 2010, on longitudinal design). Second, 2020 technology modules are single-shot, COVID-entangled, and predate large-

language-model (LLM) diffusion; they should not be read as generative-AI exposure. Occupation-linked AIOE partially time-varies exposure but was null or marginal on the Series 2 endpoints tested here. Perceived displacement is an endogenous appraisal, so associations do not separate stressor effects from selection or reverse causation, a point reinforced by the largely null AIOE. Third, workplace relatedness is a single annual item; multi-item need-satisfaction instruments were not used as primary predictors. Fourth, PsyCap overlap with profile movers is small ($n \approx 44$), and the measure is a health-framed, domain-general CPC-12 fielded in an SES-stratified subsample rather than the work-referenced Psychological Capital Questionnaire, so PsyCap pathway claims remain provisional relative to LOT-R and self-esteem even after Firth bias reduction.

Fifth, LISS entropy (.537) is lower than the reference-panel solution (about .71). Soft-weighted and pseudo-class sensitivities bound classification error for Hypothesis 4, and the sequence-based three-step estimate adjusts for it under the fixed misclassification model (Asparouhov & Muthén, 2014; Nylund-Gibson, Garber, et al., 2023; Nylund-Gibson, Grimm, et al., 2014). That exercise also surfaced a design limit worth stating: with one transition pair per person at this entropy, origin selection and transition steering are observationally near-equivalent, so identification relies on full assignment sequences and a person-mean resource, yielding an annual, trait-scaled quantity rather than an origin-year interval estimate. Hypothesis 3 moderation remains uncorrected for classification error; because classification error attenuates interaction estimates, the H3 null is conservative rather than definitive. Cohort re-enumeration makes the same point from the measurement side: the Distressed configuration is recoverable only in the pooled invariant model, so profile-level claims about Distressed exits are conditional on that measurement structure. Sixth, panel attrition is substantial (30.6% of pre-2020 entrants

observed in 2023–2025), although baseline selectivity on model variables is modest ($|d| \leq 0.18$, largest for conscientiousness) and favors healthier remainers, so late-wave estimates describe a modestly positively selected sample (Goodman & Blum, 1996).

Seventh, the economic-cost criteria are associational burden translations: the euro values inherit external multipliers (21% replacement cost; wage-based hour valuation) and wage weighting, quit intent and effort are single-wave self-reports, and no intervention or firm outcome is measured, so Hypothesis 7 supports no return-on-investment claim. Eighth, the panel spans the COVID-19 pandemic, and the 2020 stressor modules are COVID-entangled by design. Period effects were examined descriptively and by the piecewise growth check rather than adjusted by covariate: profile prevalences, within-person well-being deviations, and the formal 2020-knot model show no distress discontinuity, and prospective LISS evidence indicates the pandemic did not alter personality–mental-health associations in this panel (van der Velden et al., 2024). Designs that span the pandemic onset with repeated profile measurement (Vaziri et al., 2020) remain the stronger template where period confounding is focal; the exposure-moderation results also inherit an employed-only, post-2020 occupation frame, so exposure knot terms condition on post-knot employment. Ninth, Dutch employment protection limits generalization to U.S. midlife samples. Future work should add post-LLM exposure waves and employed-only factor-score sensitivities; the secondary heterogeneous-effect sweep run here found only weak, nonactionable subgroup structure, and repeating it on post-LLM exposure is the sharper next test.

Conclusion

The four-profile personality configuration identified in U.S. midlife adults recurred in a Dutch working-age panel observed annually across 18 years of technological change, with its

Antagonistic and Resilient anchors re-emerging within independently recruited entry cohorts. Configural structure of this kind is stable enough to serve as a common lens for studying who is vulnerable during economic transitions, and it positions the Distressed configuration as a portable marker rather than a sample artifact. Within that structure, the evidence assigns psychological resources the role conservation of resources theory predicts (Hobfoll et al., 2018). Optimism, self-esteem, and work meaning covaried with how well employees felt and, prospectively, with the direction of change among distressed employees, separating recovery from disengagement, and within persons the consequences of resource loss ran deeper and longer than the benefits of an equivalent gain. Strain tracked appraised replaceability rather than occupation-level exposure scores, which places the psychological meaning of technological change, not its occupational incidence, at the center of employee well-being during such transitions. The costs of distress also surfaced early, in lost work time and quit intent, before any realized turnover, so the window for resource-supportive action opens well before employees leave. For research, the priority is the quality of recovery from distress rather than its prevalence alone, with measured post-LLM exposure the next test of the mechanism proposed here.

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Tables

Table 1

Standardized Big Five Means for the Measurement-Invariant Four-Profile Solution (Primary Personality Waves, 2008–2025)

Profile	Share	E	A	C	N	O	MIDUS shape *r*
Distressed	16.6%	+0.38	+0.97	+0.07	+0.68	+0.61	.69
Antagonistic	28.8%	−0.76	−0.93	−0.60	+0.48	−0.60	.99
Reserved	36.3%	+0.03	−0.01	+0.01	−0.16	−0.22	.74
Resilient	18.3%	+0.80	+0.60	+0.87	−1.07	+0.83	.96

Note. N = 72,433 person-waves; entropy = .537; best log-likelihood replicated in 21 of 80 random starts. Means are within-sample z scores. Column abbreviations are the Big Five domains: E (extraversion), A (agreeableness), C (conscientiousness), N (neuroticism), and O (openness). LISS = Longitudinal Internet studies for the Social Sciences; MIDUS = Midlife in the United States study; IPIP = International Personality Item Pool (the LISS trait battery); MIDI = Midlife Development Inventory (the MIDUS trait battery). Antagonistic agreeableness is moderately low on IPIP (not MIDI-level extremity). Distressed shape r = .69 is partial.

Table 2

Hypothesis 1 Configural Correspondence Tests (LISS vs. MIDUS k = 4)

Test	Estimate	Criterion	Verdict
Mean domain linking *r*	.91	≥ .80	Supported
Neuroticism linking *r*	.78	—	Partial
Agreeableness linking *r*	.99	—	Supported
Tucker congruence (Procrustes)	.96	≥ .90	Supported
Procrustes disparity	.09	≤ .20	Supported
Distressed shape *r*	.69	—	Partial
Antagonistic shape *r*	.99	—	Supported
Reserved shape *r*	.74	—	Supported
Resilient shape *r*	.96	—	Supported

Note. Matched labels; orthogonal Procrustes on profile × trait matrices. Verdict = assessment against the prespecified criterion. Em dashes indicate profile-specific checks with no prespecified criterion; verdicts for those rows are descriptive (Partial = below the weakest prespecified benchmark). LISS = Longitudinal Internet studies for the Social Sciences; MIDUS = Midlife in the United States study.

Table 3

Hypothesis 2: Standardized Associations of 2020 AI-Related Stressors With 2025 Well-Being

Stressor (2020)	Job satisfaction (primary)	Job satisfaction (alternate)	SWLS	PA
Displacement risk	−.11*	−.11*	−.11*	.04
Exhaustion	−.22*	−.24*	−.31*	−.20*
Algorithmic management	−.03	−.03	−.10*	.03
Tech self-efficacy	.10*	.07	.10*	−.03

Note. Coefficients are standardized ordinary least squares betas with demographic covariates and heteroskedasticity-consistent (HC3) standard errors. Primary job satisfaction is the 0–10 Work-core rating; alternate is the 4-point Work item reported as robustness. Within the 16-path family, all 10 nominally significant paths survive Benjamini–Hochberg false-discovery-rate control at $q < .05$ (tech self-efficacy → 4-point job satisfaction is the nearest miss; $p = .051$). Overlap sample sizes are *n*s ≈ 787–1,406. AI = artificial intelligence; SWLS = Satisfaction With Life Scale; PA = positive affect. *$p < .05$.

Table 4

Hypothesis 4: Resources and Odds of Distressed → Resilient Versus Distressed → Reserved

Predictor	Primary *OR*	*p*	*n*	Soft *OR*	Pseudo-class	Firth *OR*	Sequence *OR*	Verdict
Optimism (LOT-R)	1.78	< .001	674	1.75	1.70 [1.48, 1.96]	1.78	2.40	Supported
Self-esteem	1.62	< .001	797	1.77	1.68 [1.47, 1.93]	1.61	3.72	Supported
Work meaning	1.77	< .001	323	1.63	1.64 [1.33, 2.02]	1.76	1.68	Supported
Relatedness	1.34	.012	332	1.14 (.227)	1.17 [0.99, 1.36]	1.33	1.64	Frame-sensitive
PsyCap total	1.41	.29	44	2.51 (.030)	2.05 [1.20, 5.19]	1.36	1.61 (.32)	Not supported

Note. Primary = pooled-leaver interval odds ratios per *SD* at the origin year (modal assignment); these are the primary Hypothesis 4 estimates. *n* = unique Distressed leavers. Recovery = Resilient; disengagement = Reserved (MIDUS-congruent contrast); MIDUS = Midlife in the United States study. Soft-weighted, pseudo-class (200 posterior draws; cells are medians with 95% intervals across draws), and Firth columns are interval-frame classification-error sensitivities. Parenthetical values in the Soft OR and Sequence OR columns are p values (*n*s = nonsignificant). Verdict = assessment against the prespecified criterion; Frame-sensitive = primary and sequence estimates significant but soft-weighted and pseudo-class sensitivities cross 1. The sequence column is corroborating only: annual latent odds ratios per *SD* of person-mean resource from the sequence-based three-step model (fixed misclassification matrix; full assignment sequences), classification-error-robust associations whose sample sizes are panel-wide (*n*s = 10,649; 12,232; 2,429; 5,602; 464), not the leaver subset; sequence magnitudes are not comparable to interval magnitudes. Sequence 95% profile CIs: optimism [1.88, 3.13]; self-esteem [2.62, 5.55]; work meaning [1.14, 2.63]; relatedness [1.26, 2.15]; PsyCap [0.59, 4.14]. *OR* = odds ratio; LOT-R = Life Orientation Test-Revised; PsyCap = psychological capital.

Table 5

Hypothesis 5: Within-Person Cross-Lagged Associations

Within-person path	β	*p*	*n* (persons)
Self-esteem$_{t-1}$ → MHI-5$_t$	−.03	< .001	7,477
MHI-5$_{t-1}$ → self-esteem$_t$	−.12	< .001	6,780
Optimism$_{t-1}$ → NA$_t$	−.06	< .001	6,748
NA$_{t-1}$ → optimism$_t$	−.02	< .001	7,432

Note. Person-mean-centered lags with lagged dependent variable; subscripts index measurement waves. MHI-5 = Mental Health Inventory-5 (scored as distress); NA = negative affect. Within-person standardized bootstrap contrasts supported a lagged asymmetry consistent with loss primacy for both pairs (self-esteem ↔ MHI-5 one-tailed $p < .001$, robust to differential reliability; optimism ↔ NA $p = .025$, fragile to differential reliability). Simultaneous random-intercept cross-lagged corroboration confirmed the asymmetry for the self-esteem–distress pair only (likelihood-ratio $p < .001$; optimism ↔ NA $p = .335$); see Figure 8.

Table 6

Hypothesis 7: Economic-Cost Contrasts for Distressed Membership

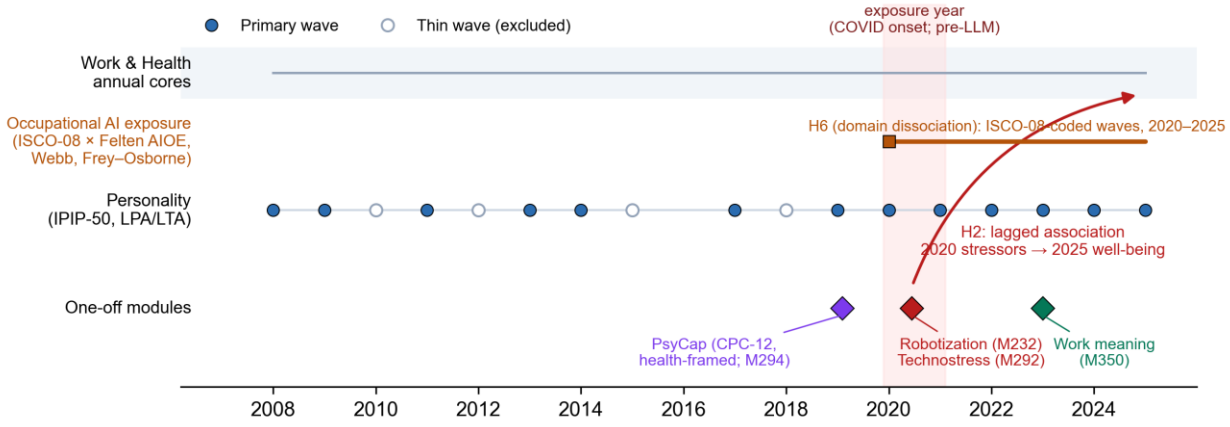
Criterion	Frame	D vs. others	*p*	D vs. Resilient	*p*	Verdict
LPT (hr per year)	2025 concurrent	+39	.078	+65	.020	Partial
Absentee days/month	2025 concurrent	+0.41	.041	+0.60	.011	Supported
LPT € (human capital)	2025 concurrent	+€691	.114	+€999	.085	Not supported
LPT € (friction cost)	2025 concurrent	+€528	.106	+€963	.020	Partial
Quit likely	2023 concurrent	+7	< .001	+9	< .001	Supported
Promotion prospects poor	2025 concurrent	+0.11	.046	+0.25	< .001	Supported
Exit paid work	2023 → 2025	+.01	.445	+.00	.937	Not supported
Realized replacement €	2023 → 2025	−€108	.251	−€265	.049	Not supported
Effort-loss €	2023 concurrent	−€2,946	< .001	−€3,422	< .001	Opposite

Note. D = Distressed; LPT = lost productive time. Estimates are ordinary least squares contrasts with heteroskedasticity-consistent (HC3) standard errors on modal assignments; 2025 frame age-adjusted for euro criteria. Verdict = assessment against the prespecified criterion; Opposite = the Distressed contrast runs opposite the predicted direction. Exit paid work entries are differences in exit proportions. Frame sample sizes: 2025 paid × Health n = 2,024 (Distressed 355), with euro and hour criteria on the subset with observed net wages and positive weekly hours (n = 1,743–1,851); 2023 → 2025 exits n = 2,213 (359); 2023 Module 350 n = 2,248 (361). Quit likely = proportion likely or very likely to quit within 12 months (Distressed 17% vs. Resilient 8%); contrasts are percentage-point differences; the scenario translation is +€330 per year expected replacement cost at 21% of salary. Euro values are burden translations under stated multipliers and wage weighting (Distressed annual net wages −€5,677 vs. Resilient, p < .001); frames differ and euros are not summed. The effort-loss criterion values the self-rated effort shortfall (100% minus reported effort) at the wage; its reversal reflects higher Distressed self-rated effort (74.5% vs. 65.9%). Double/debiased machine-learning counterweight: quit intent retained (+.05, p = .037); LPT euros attenuated (+€405, p = .35); heterogeneity tests null (ps ≥ .45).

Figures

Figure 1

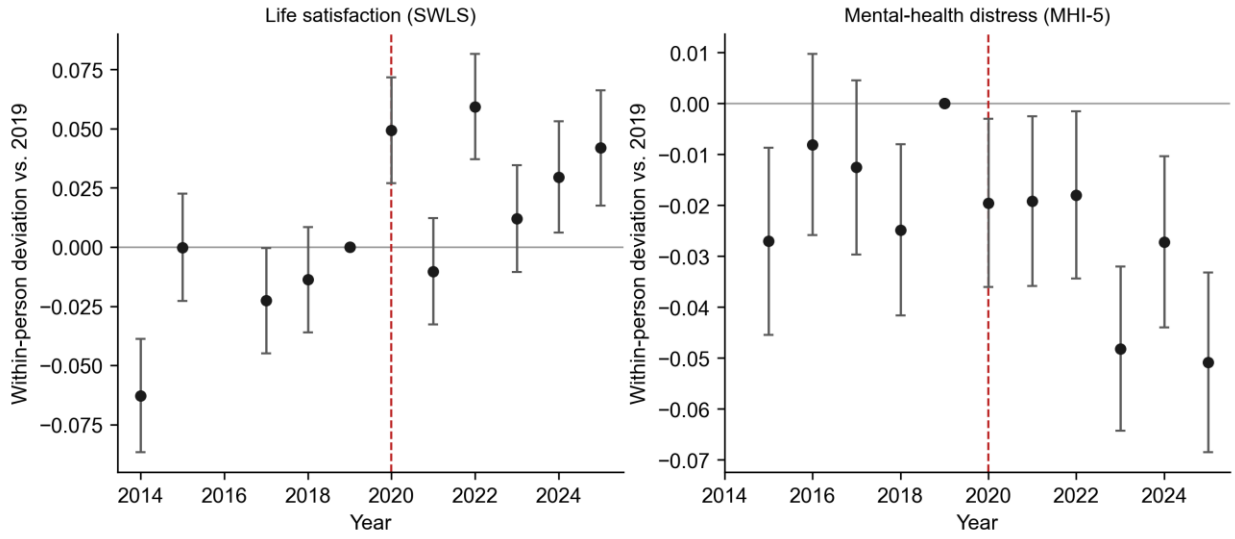
Study Design and Measurement Timing in the LISS Panel (2008–2025)



Note. Filled circles are primary Personality waves; open circles are low-completion waves (waves excluded from the primary latent transition model). Diamonds mark one-off modules: PsyCap (Module 294, February 2019; health-framed CPC-12), robotization (Module 232) and technostress (Module 292) in 2020, and work meaning (Module 350) in 2023. The orange segment marks the occupation-level AI-exposure merge, linking the AIOE (Felten et al., 2021), Webb (2020), and Frey–Osborne (Frey & Osborne, 2017) scores via ISCO-08 codes, available for Work and Schooling waves from 2020 onward and used in the Hypothesis 6 domain-dissociation test. The arrow marks the Hypothesis 2 lag from 2020 stressors to 2025 well-being criteria. Shading marks the exposure year (COVID-19 onset; before large-language-model [LLM] diffusion). LISS = Longitudinal Internet studies for the Social Sciences; IPIP-50 = International Personality Item Pool, 50-item battery; LPA/LTA = latent profile/latent transition analysis; PsyCap = psychological capital; CPC-12 = Compound PsyCap Scale; AI = artificial intelligence; AIOE = AI Occupational Exposure index; ISCO-08 = International Standard Classification of Occupations, 2008 revision.

Figure 2

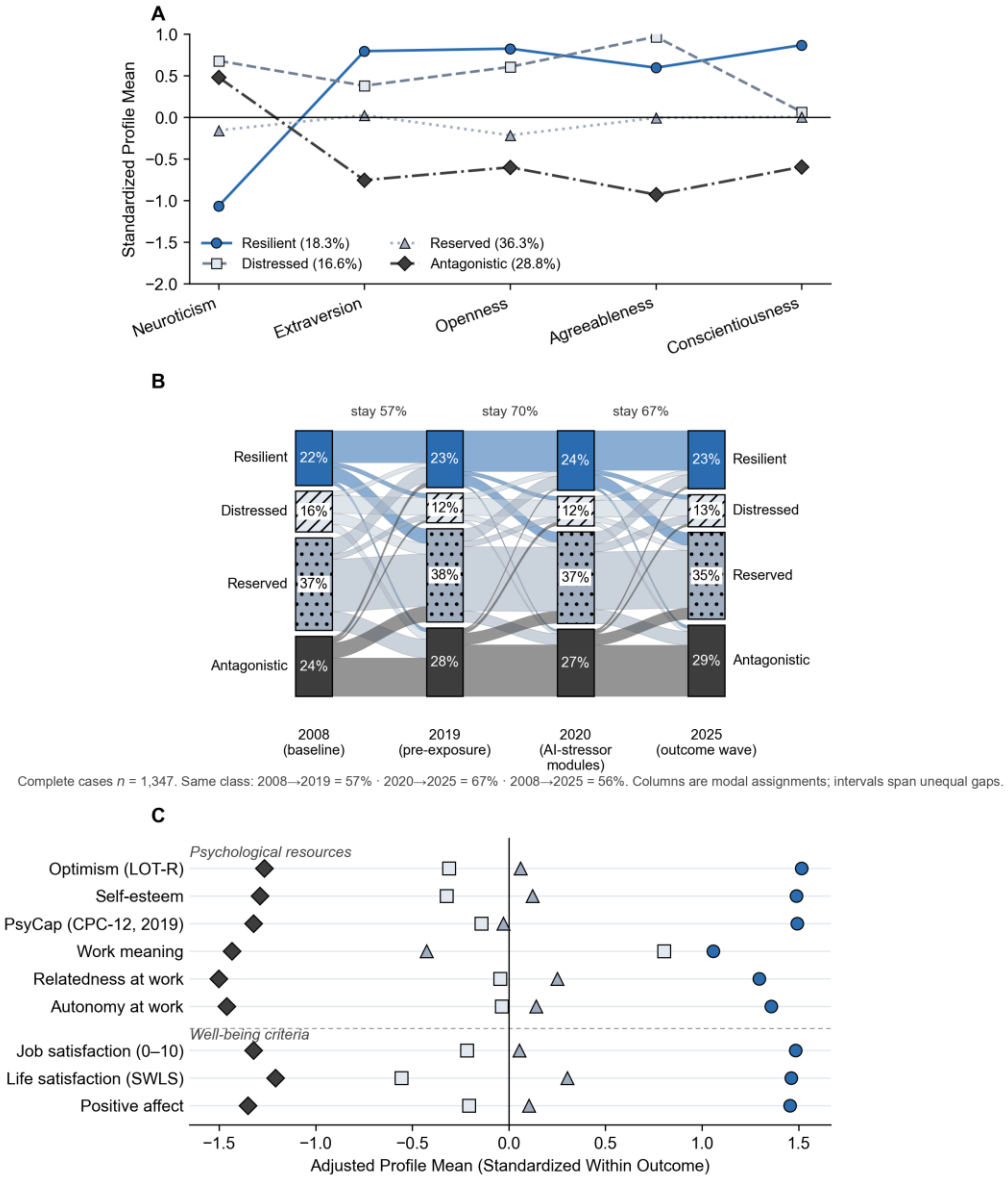
Within-Person Well-Being Deviations Around 2020 (Reference 2019)



Note. Panels show life satisfaction (SWLS) and mental-health distress (MHI-5). Points are mean person-mean-centered deviations by calendar year relative to 2019, with 95% confidence intervals, for persons with at least three observations (2014–2025); the dashed vertical line marks 2020. Descriptive, not causal: stressors were measured once (2020), so this is the population well-being series around the exposure year, not a stressor event study. SWLS = Satisfaction With Life Scale; MHI-5 = Mental Health Inventory-5.

Figure 3

Personality Profiles, Membership Dynamics, and Resource Distributions Under the Measurement-Invariant k = 4 Solution

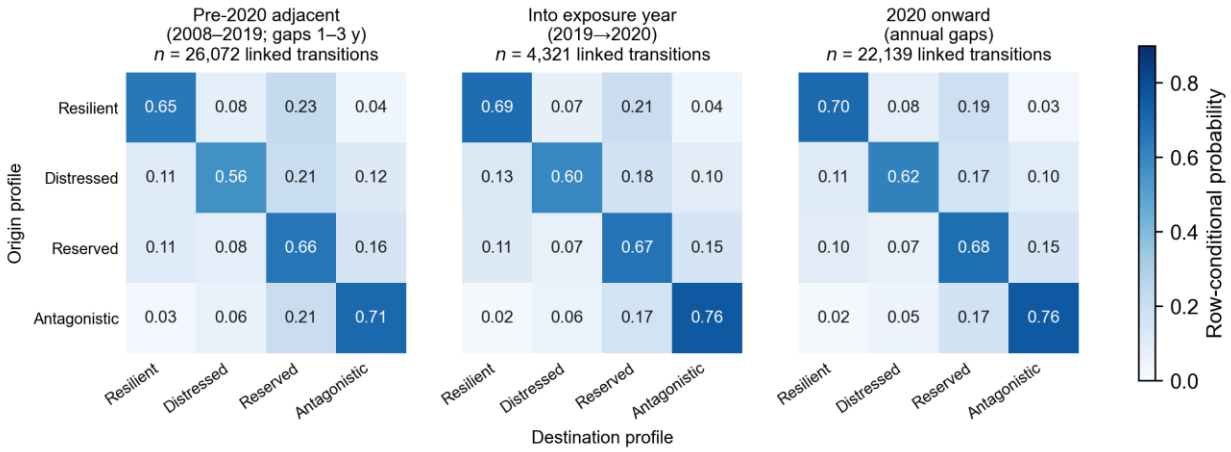


Note. Panel A: standardized Big Five profile means (primary Personality waves, 2008–2025); y-axis scaled for comparison with the MIDUS profile panel. Panel B: modal profile membership at study design anchors (2008 baseline, 2019 pre-exposure, 2020 AI-stressor module year, and 2025 outcome wave); complete-case n = 1,347; ribbon widths are transition mass and interval stay rates are printed above spans. Anchors mark measurement timing, not an interrupted time series. Panel C: age- and sex-adjusted profile means on the study's psychological resources and well-being criteria, standardized within outcome; the resource rows span the Hypothesis 4 predictors plus autonomy at work, shown as an additional resource. AI = artificial intelligence; MIDUS = Midlife in the United

States study; LOT-R = Life Orientation Test-Revised; PsyCap = psychological capital; CPC-12 = Compound PsyCap Scale; SWLS = Satisfaction With Life Scale.

Figure 4

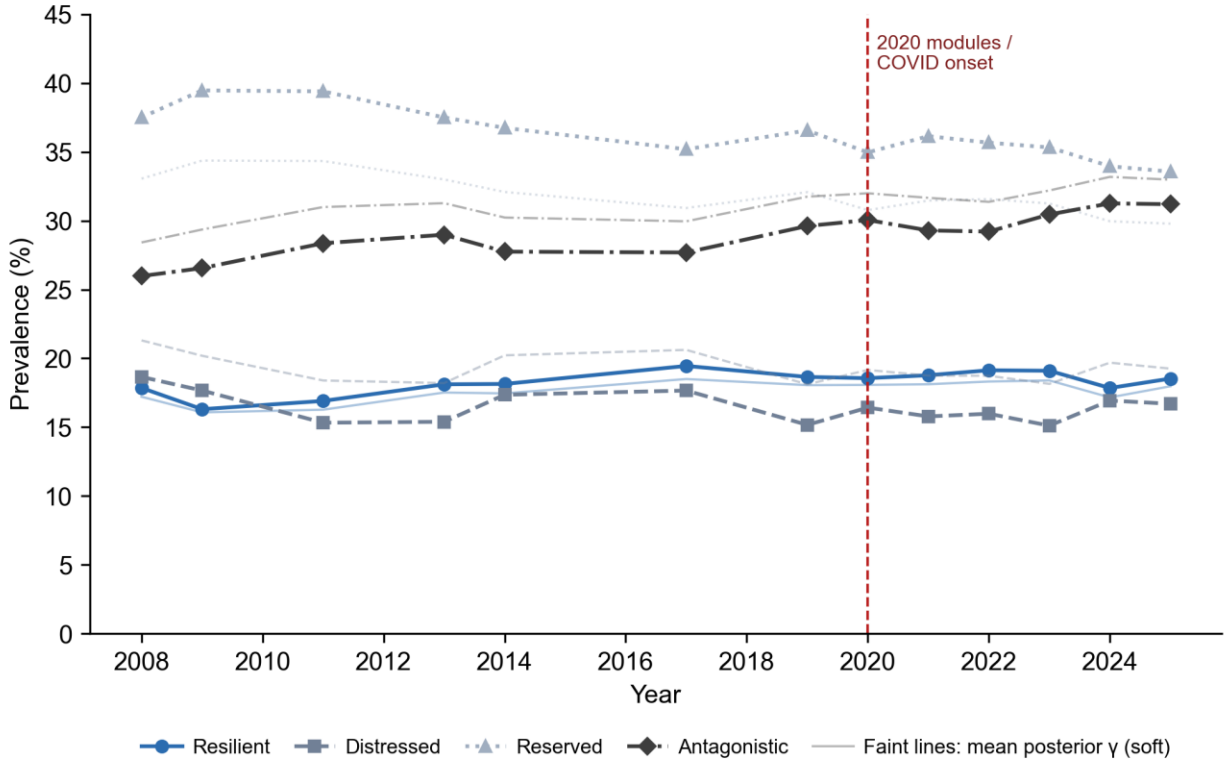
Adjacent-Wave Transition Structure Before and After 2020



Note. Cells are row-conditional probabilities of destination profile given origin profile; each panel prints its pooled count of linked transitions. The left panel pools adjacent primary-wave pairs within 2008–2019 (gaps of 1–3 years), the middle panel shows the single 2019 → 2020 transition into the exposure year, and the right panel pools annual transitions from 2020 onward. Diagonals in the pooled pre-2020 panel are not directly comparable with the annual panels without gap adjustment.

Figure 5

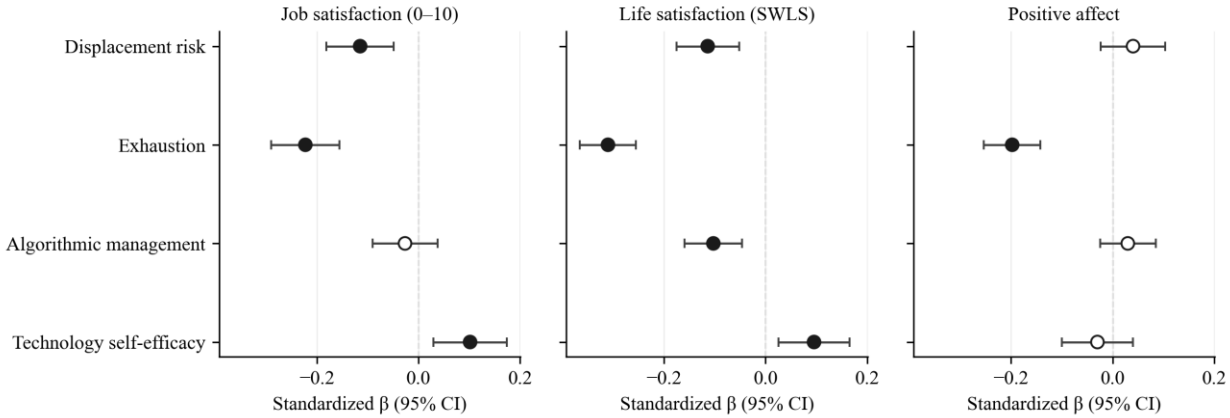
Latent Profile Prevalence Over Calendar Time



Note. Bold lines are modal-assignment shares under the measurement-invariant $k = 4$ solution; faint lines are mean posterior responsibilities (γ), the soft-classification counterweight. The dashed line marks 2020 (AI-stressor modules; COVID-19 onset). AI = artificial intelligence.

Figure 6

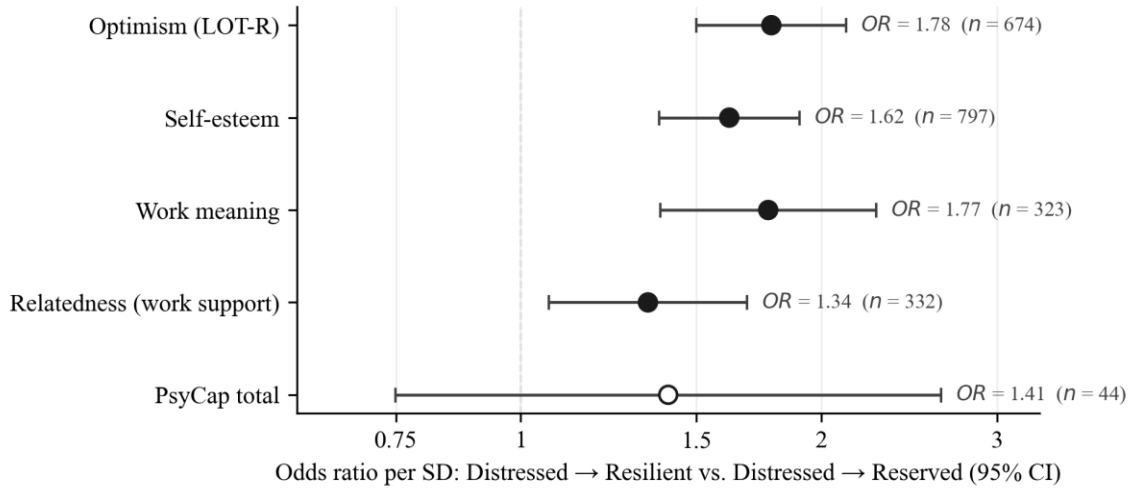
Lagged Associations of 2020 Technology-Related Stressors With 2025 Well-Being (Hypothesis 2)



Note. Points are standardized ordinary least squares betas with demographic covariates and heteroskedasticity-consistent (HC3) standard errors; bars are 95% confidence intervals. Filled markers are significant at $p < .05$. Overlap sample sizes are *n*s ≈ 799–1,430. SWLS = Satisfaction With Life Scale.

Figure 7

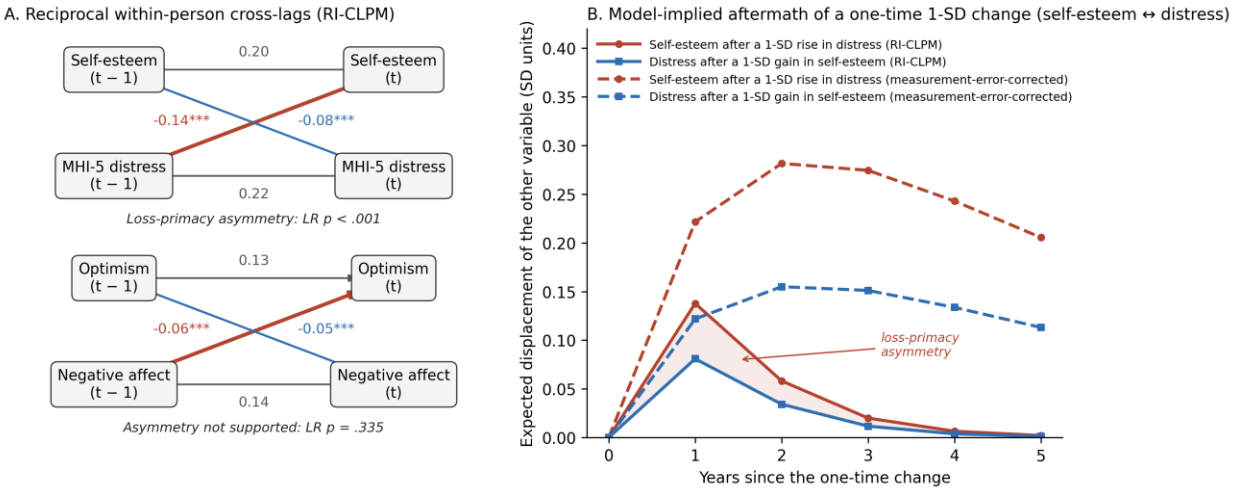
Psychological Resources and the Direction of Exit From the Distressed Profile (Hypothesis 4)



Note. Odds ratios per *SD* of each resource among pooled unique Distressed leavers, contrasting recovery-oriented (Distressed → Resilient) against disengagement-oriented (Distressed → Reserved) transitions; bars are 95% confidence intervals (log scale). Filled markers are significant at $p < .05$. The PsyCap interval reflects the small module overlap ($n = 44$). Relatedness is fragile under interval-frame classification-error sensitivities (see Table 4). *OR* = odds ratio; LOT-R = Life Orientation Test-Revised; PsyCap = psychological capital.

Figure 8

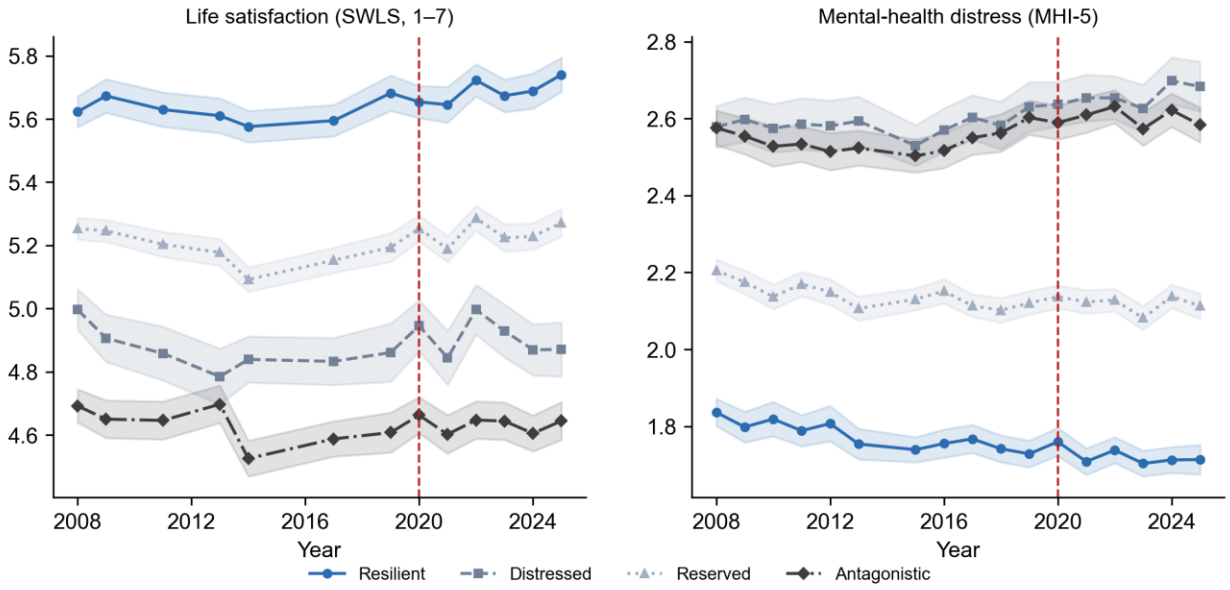
Within-Person Resource–Distress Spirals: Cross-Lagged Paths and Responses to a One-Time Within-Person Change (Hypothesis 5)



Note. Panel A: random-intercept cross-lagged panel model (RI-CLPM) estimates on pooled-standardized annual series (2017–2025) with year means removed; random intercepts are estimated but omitted from the drawing. Reciprocal cross-lagged paths of this form are a common panel operationalization of gain- and loss-spiral potential (e.g., Hakanen et al., 2008). Red paths are distress → resource (loss) cross-lags, blue paths resource → distress (gain) cross-lags, and grey horizontal paths are the within-person autoregressive coefficients. Panel B translates those paths into their multi-year consequence: each curve traces the expected displacement of one variable, in pooled-*SD* units, in the years after a one-time 1-*SD* within-person disturbance to the other variable, with all else at expected levels. Solid lines are the random-intercept cross-lagged model; dashed lines are the measurement-error-corrected latent autoregressive model (free observation noise), which separates measurement error from dynamics. The shaded region marks the loss-primacy asymmetry: a one-time rise in distress displaces self-esteem more deeply and durably than an equivalent gain in self-esteem displaces distress. The asymmetry annotations beneath each pair are likelihood-ratio (LR) tests of equal loss and gain cross-lags (self-esteem ↔ distress $p < .001$; optimism ↔ negative affect $p = .335$). MHI-5 = Mental Health Inventory-5. $^{***}p < .001$ (tests of the individual cross-lagged coefficients).

Figure 9

Life-Satisfaction and Distress Trajectories by Latent Profile (2008–2025)



Note. Points are yearly means with 95% confidence bands (cells with $n \geq 50$). SWLS uses concurrent profile assignment; MHI-5 (annual Health core) uses each person's most recent assignment within the prior 3 years. MHI-5 distress is a Hypothesis 5 and 6 endpoint rather than one of the three evaluative well-being criteria (job satisfaction, SWLS, and positive affect) used in Hypothesis 2 and the resource–well-being validity check and in Figure 3, Panel C; it is plotted here for descriptive context. The dashed line marks 2020. SWLS = Satisfaction With Life Scale; MHI-5 = Mental Health Inventory-5.

Appendix

Data Source and Prior Uses of the LISS Panel

Project. LISS panel (Longitudinal Internet studies for the Social Sciences), a probability-based household panel of the Dutch population with annual core studies and one-off modules (Scherpenzeel & Das, 2011).

Collecting entity. Centerdata, a non-profit research institute at Tilburg University, the Netherlands. Panel recruitment, informed consent, fieldwork, and data curation are Centerdata's responsibility.

Project website. https://www.lissdata.nl. Registered researchers may request the respondent-level files through the LISS Data Archive under a data-user agreement; the microdata are not redistributed.

Files used. Core Personality (2008–2025), Core Health (annual), Core Work and Schooling (annual, including ISCO-08 occupation codes), Core Background Variables, and the one-off modules Resilience towards Robotization (Module 232, 2020), Survey Technostress (Module 292, 2020), Psychological Capital (Module 294, 2019), and the 2023 work-meaning module (Module 350). Occupation-level AI exposure scores (Felten et al., 2021; Frey & Osborne, 2017; Webb, 2020) were merged by ISCO-08 code.

Other manuscripts using these data. The LISS panel has supported a large published literature. Studies cited in this article that analyze LISS data include Asselmann et al. (2021), who retained a two-class resilient versus nonresilient personality solution from the Core Personality study; van der Velden et al. (2024), who examined Big Five associations with mental health before and during the COVID-19 pandemic; and Jansen et al. (2026), who used the robotization module to

study automation risk and retraining. None of these studies estimated the profile-transition, stressor, spiral, or cost models reported here.

Data transparency. This article is the author's first use of the LISS panel. No other manuscript by the author, published, in press, under review, or in preparation, draws on LISS data or on any of the variables analyzed here, so no data-transparency table of overlapping uses is required. The author's prior profile work (Monnot, 2026) analyzed a different dataset (MIDUS) and shares no data with the present article.

Supplemental Materials

Tables S1–S3 report the exploratory stratified stressor slopes, resource–well-being validity-check paths, and economic-cost sensitivity variants referenced in the main text. Figure S1 reports participant flow and analysis-frame sample sizes.

Table S1

Exploratory Stratified Stressor Slopes (Modal Assignment)

Path	Antagonistic β	Resilient β	Distressed β
Displacement risk → job satisfaction	−.25*	−.08	+.07 (*ns*)

Note. Modal-class simple slopes (cell *n*s ≈ 160–400); descriptive only, not a substitute confirmatory moderation test. The displayed contrast is representative: across the stressor × criterion grid, stratified slopes identified Antagonistic membership, not Distressed membership, as having the steeper negative stressor slopes, consistent with companion models in which displacement risk is associated with Antagonistic membership. *$p < .05$.

Table S2

Validity Check: Psychological Resources and Well-Being

Resource	Job satisfaction β (*n*)	SWLS β (*n*)	PA β (*n*)
PsyCap total	— (79)	.45* (239)	.44* (238)
LOT-R optimism	.25* (2,149)	.52* (4,481)	.29* (4,481)
Self-esteem	.27* (2,156)	.53* (4,505)	.35* (4,484)
Work meaning	.30* (1,481)	.21* (1,691)	.20* (1,679)
Relatedness (work support)	.38* (2,213)	.20* (2,436)	.12* (2,428)

Note. Cell entries are standardized ordinary least squares betas with demographic covariates and heteroskedasticity-consistent (HC3) standard errors, with sample sizes in parentheses; all displayed paths survive Benjamini–Hochberg control ($q < .05$). The em dash marks the PsyCap → job-satisfaction overlap, which fell below the 80-case estimation floor and was not estimated. PsyCap from Module 294 (2019); work meaning from Module 350 (2023); optimism, self-esteem, and relatedness from annual cores. LOT-R = Life Orientation Test-Revised; PsyCap = psychological capital; SWLS = Satisfaction With Life Scale; PA = positive affect. *$p < .05$.

Table S3

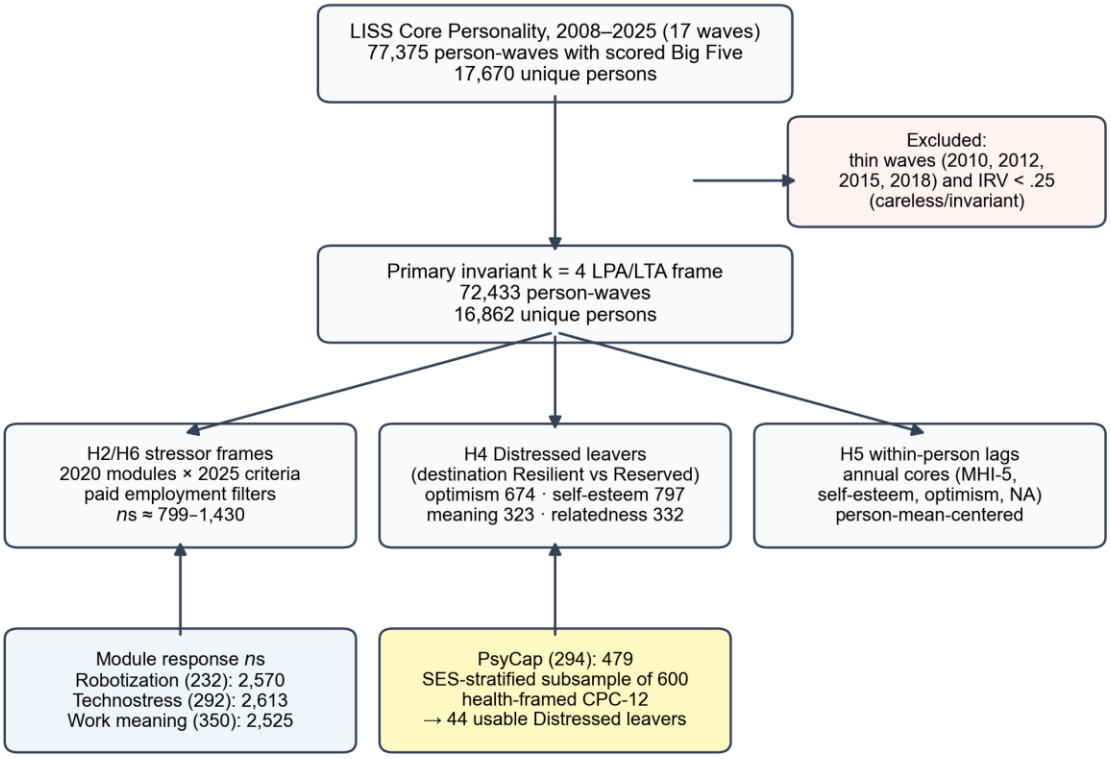
Hypothesis 7: Wage-Sensitivity and Scenario Variants

Sensitivity variant	Distressed contrast	*p*
Friction-cost LPT euros (85-day cap), versus Resilient	+€963	.020
Human-capital LPT euros, Distressed versus Resilient means	€3,166 vs. €2,167	.085
Realized replacement-cost scenario (2023 → 2025 exits)	−€265	.049
Wage-weighted effort gap, versus Resilient	−€3,422	—
Net annual wage gap, versus Resilient	−€5,677	< .001

Note. Scenario translations under stated multipliers, not measured firm outcomes. The em dash indicates that no separate significance test was run (the wage-weighted effort gap is descriptive). The realized scenario runs opposite the intent scenario because Distressed leavers earned less; the effort translation reverses because Distressed members reported higher self-rated effort (74.5% vs. 65.9%, $p < .001$). LPT = lost productive time.

Figure S1

Participant Flow and Analysis Frames



Note. Person-wave and person counts for the primary invariant LPA/LTA frame, with hypothesis-specific overlap and module-response sample sizes. The highlighted box marks the PsyCap module path (SES-stratified subsample of 600; 479 respondents; 44 usable Distressed leavers). LISS = Longitudinal Internet studies for the Social Sciences; H2, H4, H5, and H6 = Hypotheses 2, 4, 5, and 6; LPA/LTA = latent profile/latent transition analysis; IRV = intraindividual response variability (careless-responding screen); PsyCap = psychological capital; CPC-12 = Compound PsyCap Scale; SES = socioeconomic status; MHI-5 = Mental Health Inventory-5; NA = negative affect.